

Chetan Kishor C G

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EDUCATION

JSS Academy of Technical Education
B.E. Computer Science and Engineering

Bengaluru, India
Expected Jul 2027

Relevant Coursework: Database Management Systems, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, Go, Rust, TypeScript, SQL, C++

Backend: FastAPI, Go (Standard Library), Rust (PyO3), REST APIs, JWT, Redis Caching

Databases: PostgreSQL (Normalization & Schema Design), Supabase, SQLAlchemy, sqlc, pgx

DevOps & Tools: Docker, Docker Compose, k6 Load Testing, Makefile, uv, GitHub Actions

Concepts: Clean Architecture, RESTful API Design, JWT Auth, N+1 Optimization

PROJECTS

AcaTrack — Academic Analytics Platform

Jul 2025 – Present

GitHub: github.com/chetanuchiha16/acatrack | *Live:* acatrack.vercel.app

FastAPI, React, PostgreSQL

Eradicated critical **N+1 database queries** and normalized flat Excel structures into a **3NF PostgreSQL schema**, increasing load-test throughput from **1.84 to 59.6 RPS** while ensuring absolute referential integrity.

Migrated backend to an asynchronous **FastAPI** model, slashing P95 response latencies of core endpoints (e.g., student listing dropped from **343ms to 3.5ms**) under concurrent load by eliminating thread queuing.

Built a native multi-threaded **Rust engine using PyO3**, reducing academic PDF grade extraction times from **21 minutes to 1.8 minutes**.

Integrated a **Redis caching layer** for heavy student analytics and chart generation endpoints, reducing average response latency from **2.8 seconds to under 160 milliseconds**.

Developed a responsive **React** frontend with **Zustand** state stores, ensuring compile-time API validation using generated **HeyAPI** client SDKs.

IrisBridge — High-Performance Image Bridge

Jan 2026 – Present

GitHub: github.com/chetanuchiha16/iris-bridge

Rust, Python, Rayon

Developed a high-performance image processing library in **Rust** with **Python bindings via PyO3** for ML data pipelines.

Implemented **parallel batch processing** using Rayon's `par_iter`, enabling multi-core utilization for image resizing and normalization.

Optimized memory usage by returning processed images as **NumPy ndarrays** directly to Python, minimizing serialization overhead.

Go Production Blueprint (Clean Architecture)

Jan 2026 – Present

GitHub: github.com/chetanuchiha16/go-play

Go, sqlc, PostgreSQL

Architected a scalable Go backend using **Clean Architecture** principles (Handler/Service/Repository) for strict separation of concerns.

Integrated **sqlc** for generating type-safe Go code from SQL and **pgx** for efficient PostgreSQL connection pooling.

Streamlined development workflow using **Docker Compose**, **Makefile**, and **Air** for automated live-reloading and testing.

ACHIEVEMENTS

Solved **130+ LeetCode problems** focusing on sliding window, two-pointer, linked list, trees, tries, backtracking, and hashing techniques.